Omar Ibrahim

Junior Penetration Tester

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PROFESSIONAL SUMMARY

Driven Junior Penetration Tester with hands-on experience in manual web application security testing, vulnerability research, and embedded systems. Demonstrated capability in identifying real-world high-impact vulnerabilities on production systems, including Insecure Direct Object Reference (IDOR) flaws leading to unauthorized sensitive data exposure. Combines a solid technical foundation in C, Python, and Assembly with a disciplined manual-testing-first methodology to produce actionable, remediation-focused reports. Eager to contribute to offensive security operations and accelerate OSCP certification.

CORE COMPETENCIES & SKILLS

Offensive Security & Web Testing: OWASP Top 10 (IDOR, Broken Access Control, SQLi, XSS, SSRF), Manual Testing Methodology, Web Application Enumeration & Exploitation, API Security, Logic Flaw Analysis.
Security Tools: Burp Suite Professional/Community, Nmap, Metasploit Framework, SQLmap, Gobuster/FFUF, OpenVAS, Nessus, Wireshark, IDA Pro.
Programming & Automation: Python (Security Tooling/Scripts), C, Bash, PowerShell, JavaScript, SQL, x86-64 Assembly (Reverse Engineering Basics).
Systems & Platforms: Linux (Kali/Ubuntu), Windows, Git/GitHub, MySQL, Oracle SQL, Embedded Platforms (ESP32/ESP8266).

INDEPENDENT VULNERABILITY RESEARCH & PRACTICAL EXPERIENCE

Independent Web Application Security Research | *2025 – Present*

- **Production IDOR Discovery & Data Exposure**: Discovered and successfully demonstrated a critical Insecure Direct Object Reference (IDOR) vulnerability on a live production website, proving unauthorized exposure of user data via object identifier manipulation.
- **Responsible Disclosure & Reporting**: Prepared a clear, professional assessment report detailing reproduction steps, impact analysis, and specific remediation guidelines following responsible disclosure practices.

Penetration Testing & Application Security Trainee – *NTI (Cairo, Egypt)* *Aug 2026 – Sep 2026*

- Conducted manual and automated vulnerability assessments, web application testing, and WordPress hardening in accordance with industry methodologies.
- Participated in threat hunting, log analysis, and real-world capstone security assessments.

Embedded Software Developer *Aug 2023 – Jun 2025*

- Implemented Role-Based Access Control (RBAC) and least-privilege principles across industrial applications.
- Designed RFID authentication mechanisms for time-limited physical access controllers, securing board-to-hardware communication protocol interfaces.

Python Developer Intern *Oct 2025 – Nov 2025*

- Developed custom automation scripts using Regex, OS, and Shutil modules to optimize reconnaissance and security workflow tooling.

Embedded Software Developer – LABstar (Cairo, Egypt) *Aug 2023 – Jun 2025*

- Implemented Role-Based Access Control (RBAC) across multi-user industrial applications, applying least-privilege principles
- Built RFID authentication for time-limited floor access on elevator systems, including secure hardware-to-board communication
- Created a cross-platform IoT controller (Flutter + ESP32 over Wi-Fi) with real-time hardware sync and low-latency commands
- Developed inventory and sales systems on SQL databases with automated audit trails

SECURITY PROJECTS & COMPETITIONS

- **TryHackMe (Top 8% Globally):** Completed extensive practical labs covering web application penetration testing, privilege escalation, network enumeration, and CTF challenges.
- **HITU IEEE CTF (4th Place):** Competed solo against established teams of 3–5 members, solving web exploitation, cryptography, and binary analysis challenges.
- **Smart Elevator Access Controller:** Engineered hardware security logic featuring RFID authentication and access authorization rules on ESP32/Flutter platforms.
- **KeyGarden & Security Tooling (Python):** Developed logic-driven tools for secure credential generation and local security workflow automation.

PROJECTS

Bank Management System Python, CustomTkinter
- Multi-user GUI with RBAC for Customers, Tellers and Managers, plus transaction logging for accountability

Electronics Management System – LABstar SQL, RBAC
- RBAC-secured multi-user inventory system that blocks unauthorized data access and keeps audit trails

Smart Home with Voice Control (Graduation Project) Flutter, Firebase, PCB
- Voice-controlled smart home app; led the team on task distribution, PCB design and documentation

Real-Time Sensor Data Pipeline Python, SQLite, Matplotlib
- Serial data ingestion, SQLite storage and live visualization, applicable to monitoring and anomaly detection

KeyGarden – Password Generator Python, CustomTkinter
- Generator with auto, text-based, passphrase and batch export modes

Morse Code Converter x86-64 Assembly
- Win32 API only: O(1) lookup table, manual memory management and direct console I/O (low-level groundwork for reverse engineering)

Cinema Management System C
- File-based movie, ticket and seat reservation system built with modular programming

EDUCATION & CERTIFICATIONS

B.Tech in Information Technology – *New Cairo Technological University* *Expected 2029*
High School Diploma in Embedded Systems & AI (*MICA) – Ranked 3rd Nationally in Egypt* *Jun 2025*

Certifications

- Cybersecurity Analyst Professional Certificate (14 courses) **– *IBM***
- Ethical Hacking with OpenVAS & Metasploit **– *IBM***
- Database Programming with SQL **– *Oracle***

LEADERSHIP & ROLES

- **Technical Project Lead**: Led a multidisciplinary engineering team for a hardware/software smart system project.
- **Technical Coordinator (RoboCup 2025):** Managed technical rules compliance, equipment inspection, and mentoring for competing teams.